

TITLE PAGE

- **Problem Statement ID – SIH25044**
- **Problem Statement Title-** AI-Powered Crop Yield Prediction and Optimization
- **Theme-** Agriculture, FoodTech & Rural Development
- **PS Category-** Software/Hardware
- **Team ID-**
- **Team Name (Registered on portal) – Team CropLink**

Our Solution:

- ❖ **Machine Learning-driven yield prediction** using soil, weather, and historical crop data.
- ❖ Designed for **scalability** with cloud-ready architecture (future migration to AWS/GCP/Azure).
- ❖ **Optimization engine** recommending best sowing time, irrigation scheduling, and fertilizer use.
- ❖ **What-if simulation feature** to forecast yield impact under changing climate scenarios.

Unique Value Propositions (UVP):

- **Dual benefit:** Yield prediction + farming optimization (most apps stop at prediction).
- **“What-if scenario” simulation** (impact of rainfall/temp changes on yield).
- **Scalable & lightweight:** Serverless deployment via Supabase.
- **Farmer-first design:** Multi-language, offline accessibility planned.

Problem Resolution :

- ❖ Enables farmers to make **data-driven decisions**, reducing uncertainty and losses.
- ❖ Optimizes **resource use (water, fertilizer, energy)**, improving sustainability.
- ❖ Increases **productivity and farmer income** while supporting food security.
- ❖ Provides policymakers with **predictive insights** for agricultural planning.

TECHNICAL APPROACH



Programming Languages & Frameworks:

Frontend: React, TypeScript, Vite, Tailwind CSS, shadcn-ui (for dashboard UI).

Backend (future-ready): Node.js/Express.js (API layer), Cloud-hosted (AWS/GCP/Azure).

Database: PostgreSQL / MongoDB (scalable data storage).

AI/ML Models: Random Forest & XGBoost – Machine Learning algorithms for accurate yield prediction.

Visualization: Chart.js / Recharts (yield analytics & reports).

Cloud Services (Future Scalability):

AWS / GCP / Azure – For large-scale deployment, storage, and A model hosting.

Optional Hardware Integration (Future Scope):

IoT Soil Sensors – To capture pH, moisture, and nutrient data in real time.

Satellite/Drone Imagery – For NDVI-based crop health monitoring.



FEASIBILITY AND VIABILITY



Analysis of Feasibility:

- **Technical Feasibility** – AI/ML models (Random Forest, XGBoost) are well-tested and can be trained on available open datasets (Govt, FAO, Kaggle).
- **Financial Feasibility** – Initial costs are minimal (open-source datasets, free-tier cloud services). Scalable to commercial use with cloud credits/partnerships.
- **Market Feasibility** – Agriculture contributes ~18% to India's GDP; farmers, agri-tech startups, and policymakers are direct beneficiaries.

Potential Challenges & Risks:

- **Data Challenges** – Regional variations, incomplete datasets, low sensor adoption.
- **Adoption Challenges** – Digital literacy among farmers, language barriers.
- **Technical Challenges** – Scaling ML models to different crop types & geographies.
- **Financial Risks** – Cloud deployment costs if scaled nationwide.



IMPACT AND BENEFITS



Potential Impact on Target Audience:

Farmers:

- Increased crop yield through AI-driven predictions.
- Reduced dependency on guesswork → data-backed farming decisions.

Government & Policy Makers:

- Region-level yield forecasting for better food supply chain planning.
- Helps design targeted subsidy and support schemes.

Agri-Tech Startups & Researchers:

- Access to predictive insights for innovation in agriculture.

Benefits of the Solution:

•Social:

- Empowers farmers with technology → inclusivity & rural upliftment.

•Economic:

- Reduces resource wastage (fertilizer, water, energy).
- Boosts national agri-productivity, supporting GDP growth.

•Environmental:

- Prevents soil degradation by avoiding overuse of chemicals.
- Helps in climate resilience by preparing farmers for uncertain weather.

RESEARCH AND REFERENCES



- Details / Links of the reference and research work
- Kaggle DataSet ,Taylor

IMPORTANT INSTRUCTIONS



Please ensure below pointers are met while submitting the Idea PPT:

1. Kindly keep the maximum slides limit up to six **(6)**. (Including the title slide)
2. Try to avoid paragraphs and post your idea in points /diagrams / Infographics /pictures
3. Keep your explanation precise and easy to understand
4. Idea should be unique and novel.
5. You can only use provided template for making the PPT without changing the idea details pointers (mentioned in previous slides).
6. You need to save the file in PDF and upload the same on portal. No PPT, Word Doc or any other format will be supported.

Note - You can delete this slide (Important Pointers) when you upload the details of your idea on SIH portal.